⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

WEEK 06

Classes, instances and methods

ថ្នាក់ វត្ថុនិងម៉ែត្រ

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

WEEK 06

This week: learn, practise, build

សប្តាហ៍នេះ រៀន អនុវត្តនិងបង្កើត

01 · Python

Classes, instances and methods
Understand the idea and trace examples.

02 · Practice

Practise the same Python concept.
Predict, change, diagnose and explain.

03 · Project

Persistent plans and line snapshots
Build, discuss and test with a peer.

One weekly component moves the final project forward.

ផ្នែកមួយរាល់សប្តាហ៍ជួយឱ្យគម្រោងចុងក្រោយរីកចម្រើន។

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Python foundations

ម៉ូឌុលនៃ Python

Classes, instances and methods
Concept → trace a small example → explain its behavior

តាមដាន self គុណលក្ខណៈរបស់វត្ថុនិងការហៅម៉ែតូដ។

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Classes, instances and returned values

Class, instance និងតម្លៃត្រឡប់

Class and instance

Rectangle defines the type; each call creates an instance.

Attributes and self

Each rectangle stores its own width and height.

Method return

area() returns a value the caller can print or test.

Instances share method definitions while keeping their own attributes.

Instance ច្រើនមានរួម method តែរក្សា attribute របស់ខ្លួន។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Objects keep data beside related behavior

វត្ថុរក្សាទិន្នន័យជាមួយឥរិយាបថពាក់ព័ន្ធ

Class: a definition

A class describes a kind of object. __init__ sets values when an instance is created.

Instance: one object

Counter(4) and Counter(2) create separate objects with separate count attributes.

Method: behavior on an object

first.doubled() uses first as self. A method can read that instance's attributes.

Understand the behavior before changing the example.

យល់ឥរិយាបថក្នុងមុនកែឧទាហរណ៍។

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Class → instance → method call

ថ្នាក់ → វត្ថុ→ ការហៅម៉ែថូដ

1

Define a class

Describe attributes and related methods.

2

Create instances

Each instance holds its own values; an alias shares one.

3

Call a method

Python supplies the receiving instance as self.

Trace which instance receives the call and which attributes its method reads.

តាមដានវត្ថុដែលទទួលការហៅ និងគុណលក្ខណៈដែលម៉ែថូដអាន។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Syntax to read and reuse

វាក្យសម្ពន្ធសម្រាប់អាន និងប្រើឡើងវិញ

Python / syntax	Meaning	When to use it
class Name:	Define a type	Group related data and behavior
self.value	One instance attribute	Read or change this object
obj.method()	Call a method	Use the receiving instance
alias = obj	Share an object	Both names refer to one instance

Python Basics · sections 18, 13, 16 · use only when you need a lookup.

Python Basics ផ្នែក 18, 13, 16៖ មើលយោងតែពេលចាំបាច់។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Two objects share a class, not their attributes

វត្ថុពីរមានថ្នាក់រួម តែគុណលក្ខណៈដាច់ដោយឡែក

```
01 class Counter:
02     def __init__(self, count):
03         self.count = count
04     def doubled(self):
05         return 2 * self.count
06 first = Counter(4)
07 second = Counter(2)
08 print(first.doubled(), second.doubled())
```

1. Expected: 8 4.
2. Trace which instance is self on each call.
3. Change first.count; inspect both objects.

Predict → execute → explain the value at each step.

ព្យាករ → ដំណើរការ → ពន្យល់តម្លៃនៅជំហានមួយៗ។

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Hands-on practice

លំហាត់អនុវត្តផ្ទាល់

Week_06.ipynb · Session 2
Predict → run → change → explain

ព្យាករ រត់ ប្តូរ និងពន្យល់លទ្ធផល។

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Your turn: practise the same Python skills

ដល់វេនអ្នក អនុវត្តជំនាញ Python ដដែល

P1–P2 · Easy

Use the approach hints. Write the code yourself in a blank cell.

P3 · Core build

Implement a complete small program. Write expected results and assertions.

P4–P5 · Optional

Apply the skill to a new case, then attempt a harder challenge if time permits.

Write your own code, predict the result, then test and explain it.

សរសេរកូដផ្ទាល់ ព្យាករលទ្ធផល ឆ្លងសាក និងពន្យល់។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Check your work before asking for help

ធ្វើការងារមុនសុំជំនួយ

A complete implementation

Write the whole operation. Do not stop at one changed character.

Independent expected results

Calculate a normal case and an edge case before running your program.

A peer can explain it

Swap driver and reviewer. Explain one line and one test.

Explain self, separate instance attributes and method calls.

ពន្យល់ self គុណលក្ខណៈដាច់ដោយឡែក និងការហៅម៉ែតូ។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Project studio

វគ្គសិក្សាគម្រោង

Persistent plans and line snapshots
Short design briefing → most time for your work

រក្សាផែនការ និងតម្លៃបន្ទាត់ទុកនិងចាប់ផ្តើមដំបូង។

The people’s task this week

កិច្ចការរបស់អ្នកប្រើនាសប្តាហ៍នេះ

1

Someone needs help

The next volunteer needs to find the plan tomorrow.

2

This week, you will

Save a plan and retrieve it after closing and reopening the database.

3

A useful result

The next volunteer can find the same saved facts.

Use the supplied starting files. Focus on one useful result and check it.

ប្រើកសារចាប់ផ្តើមដែលផ្តល់។ ផ្តោតលើលទ្ធផលមានប្រយោជន៍មួយ ហើយធ្វើតេស្ត។

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Connect a plan to its batch-line records

ភ្ជាប់ផែនការទៅកំណត់ត្រាជួរទ្រុង

Pickup plan

Stable plan ID and explicit planning status.

Offer batch

Stable batch ID with current declaration and availability.

One plan → many lines

One batch → many lines

Plan lines

Container count and historical portions-per-container snapshot.

One plan contains lines; one batch may appear in many plans.

ផែនការមួយមានជួរច្រើន; ទ្រុងមួយអាចនៅក្នុងផែនការច្រើន។

Reference: Define a class with attributes and a method

ឯកសារយោង៖ កំណត់ថ្នាក់ គុណលក្ខណៈ និងម៉ែត្រូដ

```
01 class OfferLine:
02     def __init__(self, batch_id,
03                  portions_per_container, containers):
04         self.batch_id = batch_id
05         self.portions_per_container = (
06             portions_per_container)
07         self.containers = containers
08
09     def total_portions(self):
10         return (self.portions_per_container
11                 * self.containers)
```

1. __init__ sets the instance attributes.
2. self refers to the current instance.
3. The method returns a calculated value.

A class groups instance data and related methods.

ថ្នាក់ប្រមូលផ្តុំយកវត្តុនិងម៉ែត្រូដពាក់ព័ន្ធ

Reference: Two instances hold separate values

ឯកសារយោង៖ វត្ថុពីរក្បាតផ្ទាល់ខ្លួនផ្ទុកតម្លៃផ្សេងគ្នា

```
from rescue_models import OfferLine

first = OfferLine("DEMO", 3, 4)
second = OfferLine("DEMO", 3, 2)
print(first.total_portions(), second.total_portions())
assert first.total_portions() == 12
assert second.total_portions() == 6
```

1. Expected planning totals: twelve and six
2. An alias is not a new independent object
3. Public mutation bypasses constructor checks

Python supplies the receiving instance as self.

Python ផ្តល់វត្ថុដែលទទួលការហៅជា self។

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SQL parameters and a returned row

ប៉ារ៉ាម៉ែត្រ SQL និងជួរឌេរដែលត្រឡប់

```
01 import sqlite3
02 db = sqlite3.connect(":memory:")
03 db.execute("CREATE TABLE plans(id TEXT, portions INT)")
04 db.execute("INSERT INTO plans VALUES (?, ?)",
05             ("DEMO-P", 12))
06 row = db.execute("SELECT portions FROM plans WHERE id=?",
07                  ("DEMO-P",)).fetchone()
08 assert row == (12,)
09 print(row)
10 db.close()
```

1. Pass values separately from SQL text
2. The one-element tuple needs its comma
3. This memory-only example is not persistence evidence

A one-column result is a tuple; no match is None.

លទ្ធផលមួយជួរឈរជា tuple; គ្មានលទ្ធផលជា None។

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Paths: current directory and resolved filename

ផ្លូវកសារ៖ ថតបច្ចុប្បន្ននិងផ្លូវពេញ

```python
from pathlib import Path

print("Working directory:", Path.cwd())
path = Path("pickup_plans.sqlite").resolve()
print("Would open:", path)
print("Exists:", path.exists())
```

1. This inspection creates no database
2. A new SQLite path may create an empty file
3. Never reset useful records just to diagnose a path

A relative filename can refer to a different file from another directory.

ផ្លូវកសារធៀបអាចចង្អុលឯកសារផ្សេង ពេលប្តូរថត។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

File persistence: commit, close and reopen

រក្សាទុកឯកសារ commit បិទ និងបើកឡើងវិញ

```
from tempfile import TemporaryDirectory
from pathlib import Path
import sqlite3
with TemporaryDirectory() as folder:
    path = Path(folder) / "plans.sqlite"
    db = sqlite3.connect(path)
    db.execute("CREATE TABLE plans(portions INT)")
    with db: db.execute("INSERT INTO plans VALUES (12)")
    db.close()
    db = sqlite3.connect(path)
    print(db.execute("SELECT * FROM plans").fetchone())
    db.close()
```

1. Expected stored result: (12,)
2. Only the newly created practice folder is removed
3. The notebook also uses a fresh reader process

The second connection reads committed file state.

ការភ្ជាប់ទីពីរអានទិន្នន័យដែលបាន commit ឯកសារ។

None: handle a missing query result

None៖ ដោះស្រាយការស្វែងរកដែលគ្មានលទ្ធផល

```
import sqlite3
db = sqlite3.connect(":memory:")
db.execute("CREATE TABLE plans(id TEXT, portions INT)")
row = db.execute("SELECT portions FROM plans WHERE id=?",
                 ("unknown",)).fetchone()
if row is None:
    print("Plan not found")
else:
    print(row[0])
db.close()
```

1. A missing row returns None.
2. Use is None to choose the branch.
3. Only index the row when one exists.

Check for None before reading a selected column.

ឆែក None មុនអានជួរឈរដែលជ្រើស។

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Your implementation TODO

កិច្ចការដែលអ្នកត្រូវសរសេរក្រុម

Supplied starting point

Schema, fixture and SQL examples are supplied. Python objects are a modelling example, not a required rewrite.

You implement / do

Save P11 with Q11: 2 × 3 and Q12: 1 × 5. Commit, close, reconnect and retrieve its header and lines; handle a missing lookup.

Check the evidence

Saved total stays 11 after Q11 unit changes 3 → 4; current recalculation becomes 13. Availability stays Q11=10, Q12=4.

Recover the same planning estimate without changing batch availability.

ទាញការប៉ាន់ប្រមាណផែនការដដែល ដោយមិនផ្លាស់ប្តូរទុនដែលមាន។

⚠ Khmer: automatic translation / ភាសាខ្មែរ បកប្រែដោយស្វ័យប្រវត្តិ

Show the component working

បង្ហាញថាផ្នែកនេះដំណើរការ

Step	What to show
1 · Locate	Identify the input, this component and its next user.
2 · Implement	Save P11 and line snapshots; reopen and retrieve.
3 · Check	Saved 11; current recalculation 13; stock stays 10/4.
4 · Explain	Show your code, its actual result and why it behaves that way.

Recover the same planning estimate without changing batch availability.

ទាញការប៉ាន់ប្រមាណផែនការដដែល ដោយមិនផ្លាស់ប្តូរទិន្នន័យដែលមាន។

Further reading in English

ឯកសារអានបន្ថែមជាភាសាអង់គ្លេស

1 Python classes and instance methods

https://docs.python.org/3/tutorial/classes.html

2 Python sqlite3 documentation

https://docs.python.org/3/library/sqlite3.html

Read one relevant section, then bring an example and a question.

អានផ្នែកដែលពាក់ព័ន្ធមួយ យកឧទាហរណ៍មួយ និងសំណួរមួយមកជាមួយ។